



NEXT-GEN SYSTEM ARCHITECTURE · V2

Multi-Modal AI Ecosystem

A five-layer agentic architecture fusing computer-vision perception, agentic retrieval, an autonomous LLM orchestrator, a CI/CD-driven delivery pipeline, and zero-trust future-proofing — built for real-world, production-scale deployment.

- CNN vision
- Agentic RAG
- LLM orchestration
- CI/CD pipeline
- Zero-trust security
- Edge compute
- Self-healing

5

Architecture layers

99.9%

Target uptime

<150ms

Edge inference latency

20+

Compliance checks

SYSTEM DESIGN

Layer breakdown

●Layer 1 — Perception

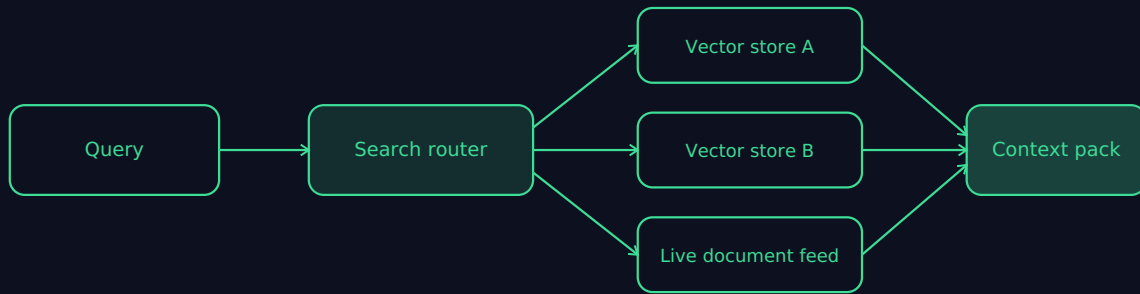
Raw sketches and sensor input are parsed into structured feature maps by a convolutional feature-extraction stack.



●Layer 2 — Agentic RAG

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A router dynamically selects across multiple vector stores rather than a static lookup, assembling grounded context for the reasoning core.



● Layer 3 — Core intelligence

The LLM orchestrator plans multi-step actions, invokes tools, and loops on its own output until the task is resolved.



● Layer 4 — Future-proofing

Zero-trust security, edge inference, and self-healing monitoring wrap the system for enterprise deployment.

Zero-trust shield

Per-request auth, encrypted transport

Edge nodes

Low-latency local inference

Self-healing loop

Health checks + auto-restart

Observability

Prometheus / Grafana telemetry

● Layer 5 — Error handling & reliability

Every request path has graceful fallbacks so failures degrade quietly instead of breaking the user experience.

Input validation

Reject malformed requests early

Retry with backoff

Transient failure recovery

Circuit breakers

Isolate failing dependencies

Structured logging

Traceable error reporting

Graceful degradation

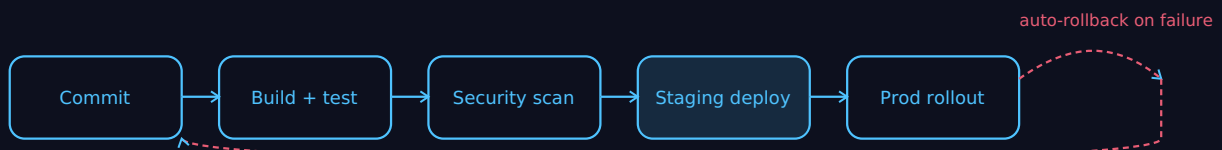
Fallback responses, cached data

Automated rollback

Bad deploys reverted automatically

DELIVERY

CI/CD pipeline



GOVERNANCE

Compliance & trust layer

Every user-facing surface of the product is checked against this list before release.

✓ Privacy policy page

✓ T&C's page

✓ Cookies policy

✓ Cookie consent check

✓ Refund policy

- ✔ **Form consent**
- ✔ **Collect only necessary data**
- ✔ **Tracking check**
- ✔ **3rd-party embeds check**
- ✔ **Accessibility fixes**
- ✔ **Alt text on images**
- ✔ **Colour contrast check**
- ✔ **Keyboard-friendly forms**
- ✔ **Clear button labels**
- ✔ **No fake reviews**
- ✔ **No false claims**
- ✔ **Business details listed**
- ✔ **Image copyright check**
- ✔ **DPDP Act compliant**
- ✔ **Other risks flagged**

*Not legal advice — have a lawyer review before production launch.

IMPLEMENTATION

Tech stack

Vision

CNN, OpenCV

Retrieval

Vector DB, LangGraph

Orchestration

LLM API, agent loop

Backend

FastAPI, Docker

Frontend

Next.js

Security

Zero-trust, encryption

Ops

Prometheus, Grafana

Deploy

GitHub Pages / Actions

WHAT'S NEXT

Roadmap

Phase 1 — Core pipeline

CNN perception + agentic RAG + LLM orchestrator live

Phase 2 — Hardening

Zero-trust security, observability, self-healing

Phase 3 — Edge rollout

Distributed edge inference nodes for low latency

Phase 4 — Compliance certification

Full DPDP / accessibility audit with legal review

CREDITS

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Multi-modal AI ecosystem — architecture overview, v2

Designed & developed end-to-end by Nikhil Chary Sriramoju. Not legal advice; have counsel review compliance items before launch.

